

Han-Teng Liao

Socio-Technical System Architect • Ex Associate Professor and Vice Dean

Taipei and Penang

✉ Email 🏠 Project 🆔 0000-0003-1081-5599 📄 Google Scholar 🌐 Web of Science 🐙 GitHub 🔗 LinkedIn 🐦 twitter

Professional Service

Peer Reviewers

Editorial

Completed 25+ reviews (mostly Q1 and Q2 journals), esp. on AI and Industry 5.0.

2021 — Present

- **Scientific Reports** (Q1:2) **Financial Innovation** (Q1:3); **Nature Humanities and Social Sciences Communications** (Q1:1)
- **Politics and Governance** (Q2:1), **Frontiers in Public Health** (Q1:1); **Journal of Clinical Epidemiology** (Q1:1)
- **Heliyon** (Q1:2); **Cogent Economics & Finance** (Q1:1); **Cogent Education** (Q2:1); **Applied Economics Letters** (Q3:1)
- **Future Internet** (Q2:1); **Discover Artificial Intelligence** (1); **Discover Education** (1)
- **Sustainability** (Q2:4); **Environment Systems and Decisions** (1)

Technical Committee Member

Technical

Joint IEEE Systems Engineering/TEMS TC on Carbon Neutrality

2026 — Present

- Explore active research and strategies to promote carbon neutrality best practices, driving consortium actions with bodies like **SiNANO** and orchestrating strategic liaisons across IEEE Technical Committees and international organizations.

Expert Contributor

International Telecommunication Union

ITU-T Focus Group on AI for Agriculture (FG-AI4A)

2023 - 2024

- Authored technical input document b-FG-AI4A-I-132 on linking carbon accounting data with agricultural business.
- Successfully integrated the framework into ITU-T Y.Suppl.83 (07/2024) under Section 7.2.7 (TG-Digital Agri-Finance).
- Translates 'Carbon Information Chains' regional best practices into international standard reference.

Student Branch Development & Launch

Guangzhou, China

IEEE Technology and Engineering Management Society (TEMS)

2019 - 2023

- Orchestrated and managed the establishment of the IEEE TEMS Student Branch during tenure as Center Director.

Education

DPhil (Ph.D.) in Information, Communication, and Social Sciences

Oxford, United Kingdom

University of Oxford

10/2007 - 10/2014

M.Sc. in Computer Science and Information Engineering

Taipei, Taiwan

National Taiwan University

01/08/2000 - 01/06/2004

M.S.S. (Master of Social Science) in Journalism

Taipei, Taiwan

National Taiwan University

01/08/2000 - 01/06/2004

B.Sc. in Electrical Engineering

Taipei, Taiwan

National Taiwan University (Includes a year-long exchange at the University of British Columbia)

01/10/1994 - 01/06/2000

Bachelor of Arts in Foreign Languages and Literatures

Taipei, Taiwan

National Taiwan University

01/08/1995 - 01/06/2000

Work Experience

Editor & Project Founder (Public Service Initiative)

Penang, Malaysia

scvcs.footprints.cv - SSbD Chips Value Chain Services | GBS.today

01/01/2026 - Current

Data Scientist and CEO

Guangzhou, China

Oxford Roadmapping

01/01/2023 - Current

Director, Higher Education Impact Assessment Center Nanfang College of Sun Yat-Sen University	<i>Guangzhou, China</i> 01/12/2019 - 01/01/2023
Associate Professor & Vice-Dean, Internet and New Media Nanfang College of Sun Yat-Sen University	<i>Guangzhou, China</i> 01/01/2017 - 01/01/2021
Research Fellow United Nations University (UNU) Institute on Computing and Society	<i>Macau, China</i> 01/07/2015 - 01/11/2016
Yahoo! Fellow, Institute for the Study of Diplomacy Georgetown University	<i>Washington DC, United States</i> 01/08/2010 - 01/07/2011
Doctoral Research Fellow Institute of Sociology, Academia Sinica	<i>Taipei, Taiwan</i> 01/08/2009 - 01/06/2010
Political Warfare Officer (Mandatory Service) Taiwan Military	<i>Taiwan</i> 01/11/2005 - 01/12/2006
Project Manager Open Source Software Foundry (OSSF), Institute of Information Science, Academia Sinica	<i>Taipei, Taiwan</i> 01/10/2004 - 30/11/2005
Action Researcher & Graduate Teaching Assistant Institute of Journalism, National Taiwan University	<i>Taipei, Taiwan</i> 01/09/2002 - 01/12/2003
Programmer & Independent Developer (Digital Publishing) Yuan-Liou Publishing Co., Ltd.	<i>Taipei, Taiwan</i> 01/07/2000 - 31/08/2000

Awards

Certificate of Merit Recognized for outstanding academic contributions	<i>IEEE TEMSMET Conference</i> 2021
Awarded for CSR (Corporate Social Responsibility) and ESG (Environmental, Social, and Governance) reporting research.	
Province-Level 'First Class' Course Curriculum Design & Pedagogy Excellence	<i>Guangdong Department of Education</i> 2020
Recognized for the development and instruction of the Capstone Project: 'API, machine learning and AI'.	
Nation-Level First Prize Instructional Excellence	<i>National College Art & Design Awards</i> 2020
Awarded for Best Teaching for Capstone Project: 'API, machine learning and AI'.	
OxTALENT Awards Best Interactive Infographics	<i>University of Oxford</i> 2014
Recognized excellence in university teaching using interactive digital content.	
Best Student Paper Award Research Excellence	<i>Chinese Internet Research Conference</i> 2009
Awarded for foundational digital community research.	
PGP Corporation and OII Scholarship Merit-based Doctoral Scholarship	<i>Oxford Internet Institute, University of Oxford</i> 2007
Value: £12,000 to support DPhil research.	
Taiwan Merit Scholarship Doctoral Research Fellowship	<i>Taiwan Government</i> 2007
Awarded for doctoral research in Digital Content (TWD \$2,000,000).	

Publications

From Stacks to Circuits: A Regenerative Socio-Technical Roadmap for AI Infrastructure within Planetary Boundaries

*ICE IEEE/ITMC, 22nd-25th June 2026, Porto
- Portugal
June 2026*

Conference Paper

- Contributors: **Han-Teng Liao**; Karen Ang
- Challenges Nvidia's "electrons-to-tokens" 5-layer scaling trajectories by converting thermodynamic entropy and Scope 3 metrics into dynamic, regeneratively accountable circuits.
- Introduces a Regenerative Socio-Technical (RST) Roadmap to address the hidden environmental, thermal, and material costs of unconstrained Generative AI scaling.
- (To be) Presented in Digital Circular Economy Special Session, providing the conceptual foundation for the **SCVCS** to anchor its AuditSoS Pillar through metrology-grounded Digital Product Passports (DPP).

Scoping Review of AI, Metrology, and ESG in the Semiconductor Sector: Implications for Safe and Sustainable by Design (SSbD)

*ICE IEEE/ITMC, 22nd-25th June 2026, Porto
- Portugal
June 2026*

Conference Paper

- Contributors: Karen Ang; **Han-Teng Liao**
- Proposes a novel 6-layer System of Systems (SoS) architecture to integrate AI-driven metrology, supply chain ESG metrics, and federated industrial data spaces into an integrated architecture.
- Bridges the gap between precision manufacturing and verifiable environmental sustainability by embedding SSbD as core operational logic for Industry 5.0.
- (To be) Presented in Technology Session, providing the conceptual foundation for the **SCVCS** to integrate RealSoS, GuardSoS and AuditSoS Pillars as a System of Systems(SoS).

Orchestrating the Twin Transition in Multinational Corporations: Technology Roadmapping for Green and Digital Global Business Services

*ICE IEEE/ITMC, 22nd-25th June 2026, Porto
- Portugal
June 2026*

Conference Paper

- Contributors: **Han-Teng Liao**; Karen Ang
- Synthesizes Technology Roadmapping (TRM) and the ITU innovation toolkit to treat Global Business Services (GBS) units as strategic "operational airlocks."
- Establishes a socio-technical framework for multinational corporations to harmonize internal digital agility and AI-native transitions with external sustainability mandates.
- (To be) Presented in Innovation Session, providing the conceptual foundation for the **SCVCS** to cross-map carbon border adjustment data aggregation under its AuditSoS Pillar for regional GBS 5.0 networks.

Collaborating on ESG consulting, reporting, and communicating education: Using partner maps for capability building design

*Frontiers in Environmental Science
March 2023*

Journal article

- Contributors: **Han-Teng Liao**; Chung-Lien Pan; Yuan Zhang
- Applied design science and circular economy frameworks to map the specific data and digital skills required for ESG consulting and reporting.
- Establishes a collaborative roadmap for universities and industry to build capability for sustainable innovation across disciplines and sectors.

Knowledge Mapping of Resilience and Human Rights in Supply Chains

*Frontiers in Environmental Science
April 2023*

Journal article

- Contributors: Chung-Lien Pan; **Han-Teng Liao** (Corresponding Author); Yuan Zhang
- Developed a roadmapping taxonomy to integrate human rights and resilience into supply chain architecture.
- Provides a governance framework for the Twin Transition, offering strategic tools for risk mitigation and value-based alignment in complex, cross-border industrial ecosystems.

Knowledge Mapping Analysis of Intelligent Ports: Research Facing Global Value Chain Challenges

Systems

- Contributors: **Han-Teng Liao**; Tsung-Ming Lo; Chung-Lien Pan
- Utilized scientometrics and science mapping to analyze the digital transformation of intelligent ports within Global Value Chains (GVC).
- Identifies technical and governance bottlenecks in open and collaborative innovation, providing a strategic roadmap for port-city digital integration.

Carbon neutrality policies and technologies: A scientometric analysis of social science disciplines'

Frontiers in Environmental Science

Journal article

October 2021

- Contributors: Yuan Zhang; Chung-Lien Pan; **Han-Teng Liao**
- Conducted a comprehensive science mapping and systematic review of 907 social science articles to delineate disciplinary boundaries and collaborative frameworks for net-zero pathways. - Highlighted the critical intersection where green digital transformation and datafied environmental indicators must guide human production and consumption patterns.

A Literature Review of Museum and Heritage on Digitization, Digitalization, and Digital Transformation'

Proceedings of the 6th International Conference on Humanities and Social Science Research (ICHSSR 2020)

Conference paper

May 2020

- Contributors: **Han-Teng Liao**; Man Zhao; Si-Pan Sun
- Executed a scientometric analysis of 1,107 Web of Science articles using VOSviewer and custom Python scripts to categorize distinct layers of socio-technical transitions. - Mapped the shift in cultural institution research fronts from technology-centric concerns to people-centric concerns (digital literacy, organizational adaptation).

Conflict and consensus in the Chinese version of Wikipedia

IEEE Technology and Society Magazine

Journal article

June 2009

- Contributors: **Han-Teng Liao**
- Investigated the socio-technical architectures, editorial policy alignments, and hyperlinked structures used to govern consensus in high-conflict, cross-border digital environments. - Analyzed how political and cultural cybernetics operate when collaborative web spaces cross sensitive jurisdictional and ideological boundaries.

More publications and citation indices are available via my ORCID Profile and Google Scholar Profile.

Academic Profiles

Impact Information

Current

- Includes comprehensive tracking of peer-reviewed articles, conference papers, and h-index metrics.

Skills

Strategic Architecture & Governance

AI System & Platform Architecture, ICT System Architecture & Analysis, Systemic Design Thinking, Software Specification Analysis, Architecture Generalization

Intelligence & Foresight (Data Science)

Strategic Foresight, Scenario Planning, Scientometrics, Bibliometrics, Science Mapping (Bibliometrix, VOSviewer), Webometrics, Knowledge Management

AI, Data Architecture & Automation

RAG (Retrieval-Augmented Generation) Context Engineering, Prompt Engineering, LLM-Driven Strategic Intelligence, Python (Pandas/NumPy), Semantic Data Modeling, Resource Description Framework (RDF), CMS Orchestration (Quarto, Obsidian, GitHub)

Product & Agile Leadership

Full-Lifecycle Product Management (MVP to Scale), PRD/MRD/BRD Documentation, UX/UI Design Research, Scrum, Agile Workflows, GitHub-Based Collaborative Engineering

Language Skills

Chinese (Mother tongue), English (Fluent — C2 across Listening, Reading, Spoken Production, Spoken Interaction, and Writing)